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Final Project Report:

Introduction:

Alzheimer's disease (AD) is a progressive brain disorder that affects memory and overall cognitive function. It starts by damaging important brain regions such as the hippocampus, which is responsible for memory, and then later spreads to the temporal and parietal lobes. As the disease progresses, communication between brain cells becomes disrupted which leads to noticeable cognitive decline.

Electroencephalography (EEG) is a non-invasive method used to record the brain's electrical activity through electrodes placed on the scalp. This is useful because it captures changes in brain activity over time. Previous research has shown that Alzheimer's disease is linked to EEG slowing. This is when there is more activity in lower-frequency brain waves (delta and theta) and less activity in higher-frequency waves (alpha and beta). These changes can appear early, which makes EEG a valuable tool for studying the disease.

In this project, we analyzed EEG data from Alzheimer's disease and Healthy Control subjects using public datasets. We extracted frequency-based features by calculating the relative power of different EEG bands (delta, theta, alpha, and beta). We then used statistical tests to check whether the differences between the two groups were significant.

To further evaluate these differences, we applied several machine learning models, including K-Nearest Neighbors (KNN), Logistic Regression, Support Vector Machine (SVM), Random Forest, and Gradient Boosting. These models were trained to classify subjects as either AD or HC based on their EEG features. This allowed us to test how well patterns in brain activity can be used to distinguish between the two groups.

Overall, this study combines signal processing and machine learning to understand how Alzheimer's disease affects brain activity. Our goal is to explore whether EEG-based features can serve as useful indicators for identifying and studying Alzheimer's disease.

Data:

The dataset used in this study consists of EEG recordings that were collected from 18 subjects, including 9 Alzheimer's disease patients and 9 Healthy Control individuals. Each subject's brain activity was recorded using 16 EEG channels placed according to the standard 10–20 electrode system, they covered key regions such as frontal, temporal, central, and parietal areas. The signals were sampled at a frequency of 256 Hz which means 256 data points were recorded per second for each channel. The recording durations varied across subjects ranging from approximately 1.5 to 9 minutes which resulted in multichannel time-series data of different lengths for each individual.

Due to the raw and high-dimensional nature of EEG signals the direct use of the data for machine learning is not effective. This is why the signals were first processed to extract meaningful features. In this project the frequency-domain features were computed by analyzing the distribution of signal power across standard EEG bands (delta, theta, alpha, and beta). These features summarize the underlying brain activity and reduce noise while preserving important patterns associated with neurological conditions. Each subject's EEG data was then transformed into a compact feature representation that could be used for statistical analysis and classification.

This step is critical because it converts complex signals into structured data that can be used by machine learning models. The resulting dataset consists of labeled feature vectors representing each subject, where the label indicates whether the subject belongs to the AD or HC group. This structured format enables both statistical comparison between groups and the application of algorithms to distinguish Alzheimer's patients from healthy individuals based on their EEG characteristics.

Machine Learning Methodology:

We trained five models on our dataset and performed hyperparameter tuning to optimize the accuracy. We chose Logistic Regression, Support Vector Machine (SVM), Random Forest, Gradient Boosting, and K Nearest Neighbors. We tuned various hyperparameters and plotted the accuracies for each model. Because our dataset only contains 18 subjects, we used Leave One Out Cross Validation (LOOCV) for model testing. This ensures that every subject is tested on and prevents overfitting. Because we used LOOCV for our models, there is no traditional train test split. One subject is used as the test set and the rest of the data is used for the train set. We rotate the test subject, so all subjects are the test set at least once.

The first model we trained on was Logistic Regression. We chose this model because it performs well on small datasets and is effective when classes are linearly separable. Our dataset only had 18 subjects and 64 features so a simpler model would have been more effective for machine learning model training. It would also help prevent overfitting. We performed hyperparameter tuning on the regularization strength C. We changed the value of C from 0.0001 to 1000 to see how it impacts the model's accuracy. When C was 0.01, we got the highest accuracy which was

88.9%. This model gave us 1 false positive and 1 false negative. The precision and recall were 0.89 for both classes. This model performed relatively well on our dataset.

The next model we trained our dataset on was SVM. SVM finds the hyperplane that best separates the 2 classes in the feature space. We chose this model because it is well suited for high dimensional and smaller datasets. We used a linear kernel because EEG band power differences for HC and AD subjects are expected to be linearly discriminable in the feature space. For hyperparameter tuning, we tested different values of C from 0.0001 to 100 to see how it would impact accuracy. We noticed that the C values did not affect model performance and consistently resulted in an accuracy of 77.7%. We tested the model on 4 kernels (linear, RBF, polynomial, and sigmoid). The accuracy was consistent at 77.7% for all kernels. The model produced 2 false positives and 2 false negatives. The precision and recall for both classes were 0.78.

The next model we implemented was Random Forest. This is an ensemble model that builds on many decision trees and aggregates their predictions. We chose this model to test whether a non-linear model can capture more complex patterns in the EEG dataset. It also provides feature importance which may help improve accuracy. We tested various n_estimator values on the model to observe how it affects accuracy. We also fixed max depth as 5. The model produced the highest accuracy at n_estimator being 50. It was able to improve model performance from 72% to 77.8%. The model's results were similar to SVM's results. However, logistic regression was able to outperform both models.

We also implemented Gradient Boosting which builds trees sequentially and corrects the errors of the previous tree. We used this model to see how our dataset performs in an iterative, error correcting approach. We tuned n_estimators from 10 to 500 to see how it impacts the model accuracy. Increasing n_estimators did not improve accuracy proving that a simpler gradient boosting model performs better on our dataset. The accuracy decreased from 66.6% to 61.1%. We also varied the learning rates from 0.001 to 0.7 The highest accuracy was 77.8% when the learning rate is 0.5. We also tested max depth from 1 to 5. We noticed that max depth did not change the accuracy, producing 77.8% accuracy across all values.

The last model we implemented was KNN which classifies a sample by looking at its k nearest neighbors. We chose to implement this model because it is non-parametric and is effective when the feature space is separated into clusters. Based on our data preprocessing steps, we noticed that AD subjects had a higher theta and delta power while HC subjects had a higher alpha and beta power. This distinction could help KNN learn the features and distinguish between HC and AD subjects. KNN achieved the highest accuracy at k equals 5. It produced a 94.4% accuracy and only misclassified one HC label.

Experiments And Evaluation:

We conducted a series of experiments to determine if EEG signals can be used to differentiate Alzheimer's Disease and Healthy patients. The first step was to extract the features. We computed the PSD (Power Spectral Density) for each subject and EEG channel. Using the Welch method, we extracted 4 EEG frequency bands.
- Delta (1-4 Hz)
- Theta (4-8 Hz)
- Alpha (8-13 Hz)
- Beta (13-30 Hz)

To ensure compatibility we calculated the relative power (Band Power/ Total Power). To ensure reliable data, we performed statistical tests to validate whether the extracted EEG bands differ between the AD and HC groups. One result is that Alpha power is lower in AD compared to HC. The p-test result is ~ 0.053, hence not statistically significant. The second result is theta power which is higher in AD compared to HC. The t-test result has a p value < 0.01 making it statistically significant. The takeaway from this is that Alzheimer's patients show EEG slowing through increased theta (slow-wave activity) and alpha (faster rhythms)

We trained multiple models to classify subjects as AD or HC. Due to the small dataset, we implemented Leave-One-Out cross-validation. We trained 17 subjects and tested one to maximize use of available data.

We evaluated the models using
- Accuracy: Correct Predictions/ Total Predictions
- Precision: Measures how many predicted AD cases are actually diagnosed with Alzheimer's
- Recall: Measures how many actual AD cases are correctly identified.
- F1 Score: Mean of precision and recall
- Confusion matrix: True vs predicted classifications

Below are the model performance comparisons listed from highest accuracy to lowest -

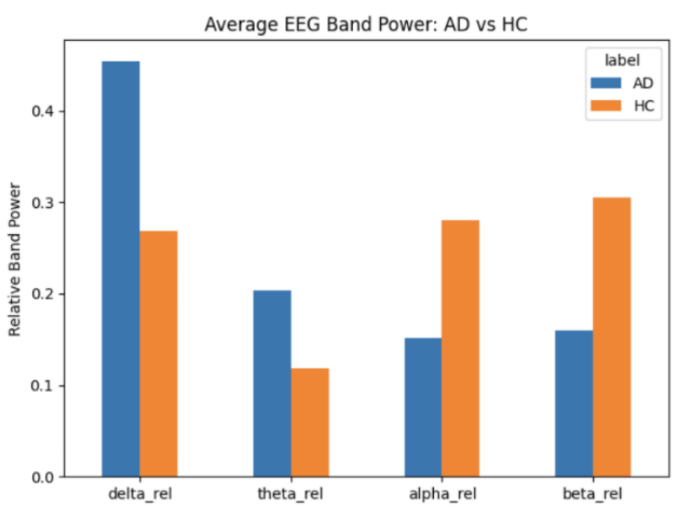
1. K-nearest neighbors: Has the highest accuracy of 94.4%. This model resulted in perfect recall (identified all AD cases with no false negatives). This model correctly classified all AD subjects and misclassified 1 HC subject. Since this is a small dataset, and the data points form clear clusters, the model produces a higher accuracy. However, this strong performance may only be due to the small dataset.
2. Logistic Regression: This model produced an accuracy of 88.9% and provided strong precision and recall. It only provided two misclassifications. This suggests that the relationship between AD and HC subjects are linear and can be separated by a simple line.
3. SVM: This model achieved an accuracy of 77.8% and showed no improvement with different kernels or regularization parameters. This suggests that increasing model complexity does not improve prediction results, hence a simpler model is sufficient.
4. Random Forest: This model achieves an accuracy of 77.8% as well. Its performance plateaued after 50 trees, showing that the complexity of the model does not improve performance. This also shows that the data does not contain complex patterns.

5. Gradient Boosting: This was the final model and performed the worst. It produced an accuracy of 61.1%. This shows poor generalization under LOO-CV. The performance did not improve even after tuning the learning rate and the number of estimators. The confusion matrix also shows that there were many misclassifications across precision and recall. This suggests that the model may be overfitting in the data instead of capturing overall patterns.

Overall conclusions from experiments:
- Models with simpler decision boundaries (KNN, Logistic Regression) performed better than the complex models.
- Recall for Alzheimer's detection was the highest in KNN with good results of both recall and precision.
- Increasing model complexity/changing parameters did not improve performance, showing that data is fully linear.
- Performance differences across models show the impact of dataset size where simple model approaches are more effective than non-linear complex ones.

Results And Discussion:

The extracted EEG frequency-based features reveal clear differences between Alzheimer's Disease (AD) and Healthy Control (HC) subjects. Consistent with prior research, AD is associated with EEG slowing, characterized by increased low-frequency activity (delta and theta) and reduced high-frequency activity (alpha and beta).



Group	Delta Power	Theta Power	Alpha Power	Beta Power
AD	0.454	0.203	0.152	0.159

HC	0.269	0.119	0.280	0.305

Figure A. Average relative EEG band power for Alzheimer's disease (AD) and Healthy Control (HC) groups.

These results from the dataset are consistent with the expected pattern of EEG slowing in Alzheimer's disease. AD subjects exhibit higher delta and theta power and lower alpha and beta power compared to HC subjects, confirming that EEG band power features capture meaningful physiological distinctions between the two groups.

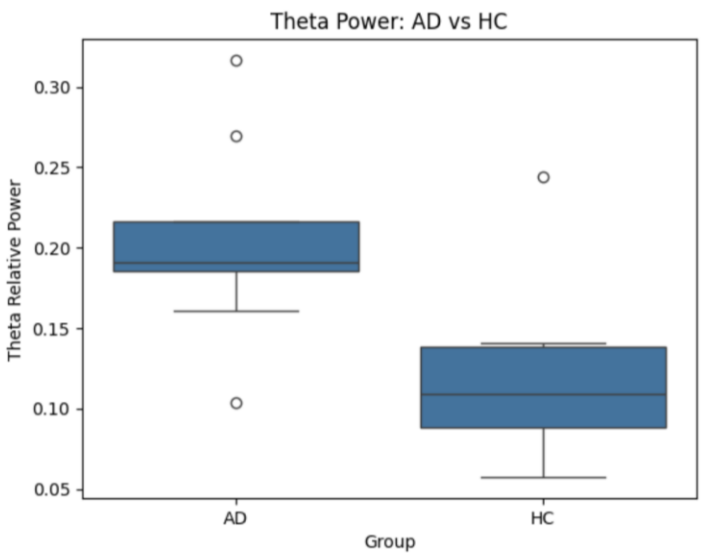


Figure 1. Boxplot of theta band power for Alzheimer's Disease (AD) and Healthy Control (HC) subjects.

As shown in Figure 1, AD subjects exhibit consistently higher theta power compared to healthy controls. Most AD values are concentrated around 0.18–0.22, whereas HC subjects fall within a lower range of approximately 0.08–0.14. This visible separation highlights the discriminative nature of theta band activity. To further evaluate this difference, a two-sample t-test was conducted. The results yielded a t-statistic of 3.12 and a p-value of 0.0066, which is below the significance threshold of 0.05, indicating that the difference in theta power between AD and HC groups is statistically significant.

Multiple machine learning models were evaluated to assess how effectively the features could classify between AD and HC subjects.

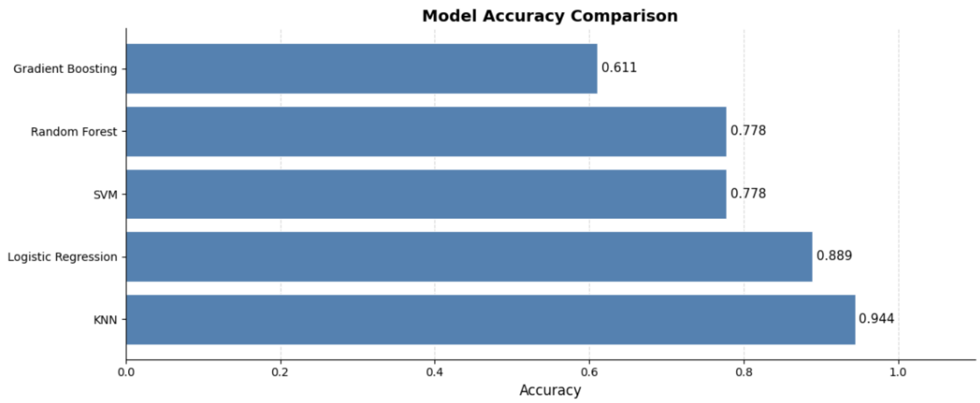


Figure 2. Model accuracy comparison across classifiers.

Among all evaluated ML models, K-Nearest Neighbors (KNN) achieved the highest accuracy (0.944), correctly classifying 17 out of 18 subjects.

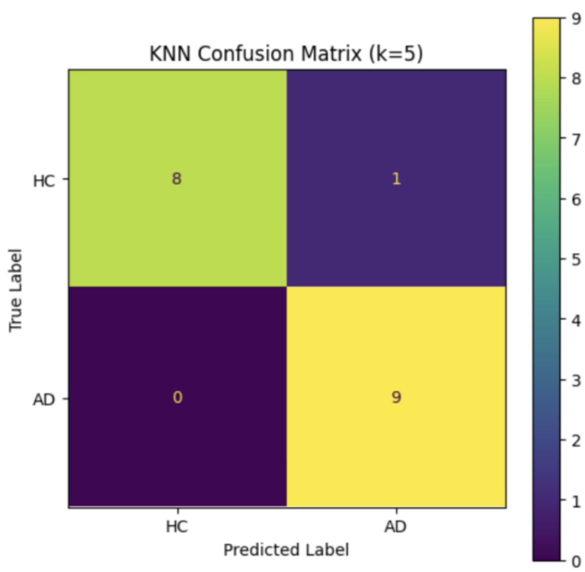


Figure 3. Confusion matrix for KNN (k=5).

The confusion matrix shows that all AD subjects were correctly identified, with only one HC subject misclassified. This suggests that the extracted EEG features form relatively distinct groupings in the feature space, allowing a distance-based model such as KNN to perform effectively.

Logistic Regression also demonstrated strong performance, achieving an accuracy of **0.889** with balanced classification across both groups. This indicates that the relationship between EEG band power features and class labels is largely linear, meaning that relatively simple models are sufficient to capture the key differences between AD and HC subjects. In contrast, Support Vector Machine (SVM) and Random Forest achieved moderate performance (accuracy of **0.778**), with more frequent misclassifications. Gradient Boosting performed the worst (accuracy of **0.611**), likely due to overfitting or instability caused by the small dataset size, where more complex models do not have sufficient data to learn reliable patterns.

From a neurological perspective, these findings align with the well-established phenomenon of EEG slowing in Alzheimer's disease. The observed increase in low-frequency activity and decrease in high-frequency activity in AD subjects help explain why machine learning models are able to distinguish between the two groups.

However, a key limitation of this study is the small sample size (18 subjects). While KNN achieved the highest accuracy, its performance may not generalize well to larger datasets, as it relies on local patterns in the data. Similarly, the effectiveness of all models may change when applied to more diverse or noisier datasets.

Overall, the results suggest that EEG-based frequency features are informative for distinguishing Alzheimer's patients from healthy individuals. However, larger datasets and more robust validation methods are needed to confirm the reliability and generalizability of these findings.

Future Work:

Although this study demonstrates that EEG based frequency features can effectively distinguish between Alzheimer's Disease (AD) and Healthy Control (HC) subjects, several improvements can be made to enhance the robustness, reliability, and clinical relevance of the approach.

One of the primary limitations is the small dataset size and the use of block level segmentation. While dividing EEG recordings into smaller blocks increases the number of training samples, it may introduce correlations between samples from the same subject. Although Leave One Out Cross Validation was applied, future work should implement subject level validation such as leave one subject out cross validation to ensure a more realistic assessment of model generalization and to prevent data leakage.

In terms of feature extraction, this project focused on relative band power across standard EEG frequency bands. While effective, this representation is relatively simple and may not fully capture the complexity of brain activity. Future work should explore more advanced features including time frequency representations such as wavelet transforms, spectral entropy, and measures of functional connectivity between EEG channels. Incorporating spatial relationships between electrodes could also improve the model's ability to capture region specific neurological patterns.

From a modeling standpoint, hyperparameter tuning was explored for select models such as Logistic Regression and Gradient Boosting. However, a more systematic and comprehensive optimization strategy such as grid search or randomized search should be applied across all models including KNN, SVM, and Random Forest. This would ensure a fair comparison and could lead to improved performance.

Future work could also investigate more advanced learning approaches. While traditional machine learning models performed well, deep learning methods such as Convolutional Neural Networks and Recurrent Neural Networks can learn complex spatial and temporal patterns directly from raw EEG signals. These approaches may reduce the need for manual feature engineering, although they require larger datasets to be effective.

In addition to accuracy, future evaluations should incorporate a broader set of performance metrics including precision, recall, F1 score, and ROC AUC to provide a more complete understanding of model performance, especially in clinically relevant scenarios.

Another important direction is improving model interpretability. Identifying which EEG features, frequency bands, and brain regions contribute most to classification decisions would make the results more meaningful and useful in a clinical context. Techniques such as feature importance analysis or interpretability methods can help achieve this.

Finally, the scope of the problem can be expanded beyond binary classification. Future studies could include intermediate conditions such as Mild Cognitive Impairment or focus on predicting disease progression over time. This would increase the clinical impact of the work by supporting earlier detection and intervention.

Overall, improving validation strategies, expanding feature representations, optimizing models, and increasing dataset size will be essential steps toward developing a more accurate and clinically useful EEG based system for Alzheimer's disease analysis.

Contribution:

- Anika Guin:
- Soujanya Prakash Kamalapur:
- Nandini Paidesetty:
- Nihita Soma: Experiments and Evaluation
- Meenakshi Sundar Rajan: